

Nils Lukas

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Updated on July 26, 2026

Research Interests

Design secure and private AI agents in the presence of untrustworthy **providers**, **data**, **models**, and **users**.

1. **Providers**: Confidential computing & encrypted inference.
2. **Data**: Poisoning, prompt injection & training-data privacy.
3. **Models**: Scalable oversight & differentially private inference.
4. **Users**: Detect misuse via watermarking & safety monitoring.

Research Grants

Awarded	\$1,458,000
[United AI-Saqer Group] MedBoard: Privacy-preserving Multi-Agent AI for Healthcare	\$136,000 · 2026
PI: Nils Lukas · Co-PIs: Praneeth Vepakomma , Preslav Nakov	
[IAAI Research Program] Trust Layer for Safe Smallholder Advisory	\$500,000 · 2026
PI: Salem Lahlou · Co-PIs: Nils Lukas, Martin Takáč	
[Amazon Research Awards – Special Call] AdvSim2Real: Robustifying Agents in Simulated Adversarial Environments	\$100,000 · 2025
PI: Nils Lukas · Co-PI: Praneeth Vepakomma	
[Etihad Airways] Conversational Booking Agents	\$450,000 · 2025
PI: Nils Lukas · Co-PIs: Salem Lahlou , Alham Fikri Aji , Martin Takáč , Mingming Gong	
[United AI-Saqer Group] Privacy-preserving Brain–Computer Interfaces	\$136,000 · 2025
PI: Abdulrahman Mahmoud · Co-PIs: Nils Lukas, Elizabeth Churchill	
[TII Funding] GFlowNets for Fuzzing of Agentic Applications	\$136,000 · 2025
PIs: Salem Lahlou & Nils Lukas	

Conference Publications

Underlined authors are students and postdocs I supervise as primary advisor.

23 [ECCV'26]	SPQR: A Multi-Dimensional Benchmark for Safety Alignment under Benign Model Adaptation Mohammed Talha Alam, Nada Saadi, Fahad Shamshad, Nils Lukas, Karthik Nandakumar, Fakhri Karray, Samuele Poppi . European Conference on Computer Vision, 2026.
22 [CVPR'26] AR: 25.4% (4,090/16,092) 🏆 Oral (top 0.8%)	RAVEN: Erasing Invisible Watermarks via Novel View Synthesis Fahad Shamshad, Nils Lukas, Karthik Nandakumar. IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026.
21 [ICML'26] AR: 26.6% (6,352/23,918)	Collaborative Threshold Watermarking Tameem Bakr , Anish Ambreth, Nils Lukas. The Forty-Third International Conference on Machine Learning, 2026.
20 [ICML'26] AR: 26.6% (6,352/23,918)	CoRe: Collaborative Reasoning via Cross Teaching Kshitij Mishra , Mirat Aubakirov, Martin Takáč, Nils Lukas, Salem Lahlou. The Forty-Third International Conference on Machine Learning, 2026.

19	Forest Before Trees: Latent Superposition for Efficient Visual Reasoning
[ACL'26]	Yubo Wang, Juntian Zhang, Yichen Wu, Yankai Lin, Nils Lukas , <u>Yuhan Liu</u> . The 64th Annual Meeting of the Association for Computational Linguistics, 2026.
Main Conference	
AR: 18.9%	
(2,296/12,148)	
18	Detecting AI-Generated Video: A Vision-Language Dual-View Survey
[ACL'26]	Dylan Xinming Hou, Juntian Zhang, Xu Gu, Yichen Wu, Nils Lukas , Gus Xia, Xiuying Chen, <u>Yuhan Liu</u> . Findings of the Association for Computational Linguistics, 2026.
Findings	
AR: 17.8%	
(2,163/12,148)	
17	DP-Fusion: Token-Level Differentially Private Inference for Large Language Models
[ICLR'26]	<u>Rushil Thareja</u> , Preslav Nakov, <u>Praneeth Vepakomma</u> , Nils Lukas . The Fourteenth International Conference on Learning Representations, 2026.
AR: 27.4%	
(5,355/19,525)	
16	SD-E²: Semantic Exploration for Reasoning Under Token Budgets
[EACL'26]	<u>Kshitij Mishra</u> , Nils Lukas , Salem Lahlou. Findings of the European Chapter of the Association for Computational Linguistics, 2026.
Findings	
15	Mask Image Watermarking
[NeurIPS'25]	Runyi Hu, Jie Zhang, Shiqian Zhao, Nils Lukas , Jiwei Li, Qing Guo, Han Qiu, Tianwei Zhang. The Thirty-Ninth Annual Conference on Neural Information Processing Systems, 2025.
AR: 24.5%	
(5,290/21,575)	
14	SPIRIT: Patching Speech Language Models against Jailbreak Attacks
[EMNLP'25]	Amir Djanibekov, Nurdaulet Mukhituly, Kentaro Inui, Hanan Aldarmaki, Nils Lukas . Empirical Methods in Natural Language Processing (Main Conference), 2025.
AR: 22.2%	
(1,811/8,172)	
13	Optimizing Adaptive Attacks against Content Watermarks for Language Models
[ICML'25]	Abdulrahman Diaa, <u>Toluwani Aremu</u> , Nils Lukas . The Forty-Second International Conference on Machine Learning, 2025.
AR: 26.9%	
(3,260/12,107)	
★ Spotlight	
(Top 2.6%)	
12	Cowpox: Towards the Immunity of VLM-based Multi-Agent Systems
[ICML'25]	Yutong Wu, Jie Zhang, Yiming Li, Chao Zhang, Qing Guo, Han Qiu, Nils Lukas , Tianwei Zhang. The Forty-Second International Conference on Machine Learning, 2025.
AR: 26.9%	
(3,260/12,107)	
11	PEPSI: Practically Efficient Private Set Intersection in the Unbalanced Setting
[USENIX'24]	Rasoul Mahdavi, Nils Lukas , Faezeh Ebrahimiaghazani, Thomas Humphries, Bailey Kacsmar, John Premkumar, Xinda Li, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024.
AR: 19.1%	
(417/2,176)	
10	Fast and Private Inference of Deep Neural Networks by Co-designing Activation Functions
[USENIX'24]	Abdulrahman Diaa, Lucas Fenaux, Thomas Humphries, Marian Dietz, Faezeh Ebrahimiaghazani, Bailey Kacsmar, Xinda Li, Nils Lukas , Rasoul Akhavan Mahdavi, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024.
AR: 19.1%	
(417/2,176)	
9	Leveraging Optimization for Adaptive Attacks on Image Watermarks
[ICLR'24]	Nils Lukas , Abdulrahman Diaa, Lucas Fenaux, Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.
AR: 30.8%	
(2,250/7,262)	

8	Universal Backdoor Attacks
[ICLR'24]	Benjamin Schneider, Nils Lukas , Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.
AR: 30.8% (2,250/7,262)	
● Media Coverage	
7	PTW: Pivotal Tuning Watermarking for Pre-Trained Image Generators
[USENIX'23]	Nils Lukas and Florian Kerschbaum. In the 32nd USENIX Security Symposium, 2023.
AR: 29.2% (422/1,444)	
6	Analyzing Leakage of Personally Identifiable Information in Language Models
[S&P'23]	Nils Lukas , Ahmed Salem, Robert Sim, Shruti Tople, Lukas Wutschitz, Santiago Zanella-Béguelin. In the 44th IEEE Symposium on Security and Privacy, 2023.
AR: 17.0% (195/1,147)	
★ Distinguished Contribution Award at Microsoft MLADS	
5	SoK: How Robust is Image Classification Deep Neural Network Watermarking?
[S&P'22]	Nils Lukas , Edward Jiang, Xinda Li, Florian Kerschbaum. In the 43rd IEEE Symposium on Security and Privacy, 2022.
AR: 14.5% (147/1,012)	
4	Deep Neural Network Fingerprinting by Conferrable Adversarial Examples
[ICLR'21]	Nils Lukas , Yuxuan Zhang, Florian Kerschbaum. The Ninth International Conference on Learning Representations, 2021.
AR: 28.7% (860/2,997)	
★ Spotlight (Top 5%)	
3	On the Robustness of Backdoor-based Watermarking in Deep Neural Networks
[IH&MMSEC'21]	Masoumeh Shafieinejad, Nils Lukas , Jiaqi Wang, Xinda Li, Florian Kerschbaum. Proceedings of the 2021 ACM Workshop on Information Hiding and Multimedia Security, 2021.
AR: 40.3% (128/318)	
2	Practical Over-Threshold Multi-Party Private Set Intersection
[ACSAC'20]	Rasoul Mahdavi, Thomas Humphries, Bailey Kacsmar, Simeon Krastnikov, Nils Lukas , John Premkumar, Masoumeh Shafieinejad, Simon Oya, Florian Kerschbaum, Erik-Oliver Blass. Annual Computer Security Applications Conference (ACSAC), 2020.
AR: 20.9% (104/497)	
1	Differentially Private Two-Party Set Operations
[EuroS&P'20]	Bailey Kacsmar, Basit Khurram, Nils Lukas , Alexander Norton, Masoumeh Shafieinejad, Zhiwei Shang, Yaser Baseri, Maryam Sepehri, Simon Oya, Florian Kerschbaum. IEEE European Symposium on Security and Privacy (EuroS&P), 2020.
AR: 20.9% (39/187)	

PostDocs, Students

Postdoctoral Researchers

- [Samuele Poppi](#) since 2025
- [Kshitij Mishra](#) (*secondary: [Martin Takáč](#) & [Salem Lahlou](#)*) since 2025
- [Dushyant Chauhan](#) (*secondary: [Martin Takáč](#) & [Salem Lahlou](#)*) since 2025
- [Yuhan Liu](#) (*secondary: [Martin Takáč](#) & [Salem Lahlou](#)*) since 2025

Research Engineers

- [Sarim Hashmi](#) since 2025

PhD Students

- [Fabian Zhang](#) (ML) since 2025
- [Tair Djanibekov](#) (NLP) since 2025
- [Rushil Thareja](#) (NLP) (*secondary: [Praneeth Vepakomma](#)*) since 2024
- [Toluwani Aremu](#) (ML) since 2023

	MSc Students	
	• Juan Nicolas Sepulveda Arias (CV)	since 2025
	• Mohamed Mahmoud Mohamed Hendy (ML)	since 2025
	• Majid Mohamed Mohamed Sulaiman Ibrahim (ML)	since 2025
	• Prince Jha (ML) (<i>primary: Salem Lahlou</i>)	since 2025
	• Muhammad Mohideen (CV) (<i>secondary: Samuel Horvath</i>)	since 2025
	Alumni	
	• Tameem Bakr (ML)	MSc · Mar 2026
	• Shaikha Jasem Mohamed Ismail Alhosani (ML)	MSc · Mar 2026
	• Maryam Abdulla Rashed Abdulla Alshamsi (ML)	MSc · Mar 2026
Talks & Keynotes	• Maryam Alshehyari (ML)	MSc · Mar 2026
	• Abdalla Khalid Mohamed Abdalla Alzaabi (ML)	MSc · Mar 2026
	• Ali Al-Ali (ML)	MSc · Mar 2026
	AI Safety Workshop: Alignment, Oversight and Deception	
	• ADIA Lab & UGR Summer School , Granada	2026
	The Role of Watermarking for ML Security	
	• CyberQ , UAE Cyber Security Council & TII, Abu Dhabi	2025
	Adaptively Robust and Forgery-Resistant Watermarking	
	• Meta (FAIR), hosted by Hady Elsahar	2025
	• International Symposium on Trustworthy Foundation Models, MBZUAI	2025
	Emerging Topics in Machine Learning	
	• Build with AI , Google Developer Groups, Abu Dhabi	2025
	Optimizing Adaptive Attacks against Content Watermarks	
	• DeepMind, hosted by David Stutz	2024
	• University of California, Berkeley, hosted by Dawn Song	2024
	Analyzing Leakage of Personal Information in Language Models	
	• Microsoft M365, hosted by Robert Sim	2024
	• Meta, hosted by Will Bullock	2023
	• MongoDB, hosted by Marilyn George and Archita Agarwal	2023
	How Reliable is Watermarking for Image Generators?	
Honors & Awards	• Google, hosted by Somesh Jha	2023
	• University of California, Berkeley, hosted by Dawn Song	2023
	Keynotes	
	• Aviation Future Week , hosted by Emirates, Dubai	2024
	• Cyber Energy Leadership Forum , Abu Dhabi	2024
	• Amazon Research Award [100,000 USD]	2025
	• First Place at the NeurIPS'24 Watermarking Competition [4,400 USD]	2024
	• First Place at DGE Elite Hackathon, GITEX'24 [10,900 USD]	2024
	• Top Mathematics Doctoral Thesis , University of Waterloo [1,080 USD]	2024
	• Alumni Gold Medal , One PhD Award Yearly, University of Waterloo	2024
Workshop Papers	• Best Poster Award, Sponsored by David R. Cheriton [220 USD]	2023
	• Distinguished Contribution Award, Microsoft MLADS conference	2023
	• David R. Cheriton Scholarship, University of Waterloo [14,400 USD]	2022, 2023
	• Outstanding Reviewer, ICML'22	2022
	• Best Poster Award, Sponsored by Rogers [720 USD]	2019
	3 Sanitizing Medical Documents with Differential Privacy using Large Language Models	
	[GenAI4Health] Rushil Thareja , Gautam Gupta , Preslav Nakov , Praneeth Vepakomma , Nils Lukas . 2025.	

	2	First-Place Solution to NeurIPS 2024 Invisible Watermark Removal Challenge
	[WMARK'25]	Fahad Shamshad, Tameem Bakr, Yahia Salaheldin Shaaban, Noor Hazim Hussein, Karthik Nandakumar and Nils Lukas . The 1st Workshop on GenAI Watermarking, 2025.
	1	Optimizing Adaptive Attacks against Content Watermarks for Language Models
	[WMARK'25]	Abdulrahman Diaa, Toluwani Aremu, Nils Lukas . The 1st Workshop on GenAI Watermarking, 2025.
	★ Oral Presentation	
Working Papers		Mitigating Watermark Forgery in Generative Models via Randomized Key Selection , Submitted.
		Toluwani Aremu, Noor Hazim Hussein, Munachiso S Nwadike, Samuele Poppi, Jie Zhang, Karthik Nandakumar, Neil Zhenqiang Gong, Nils Lukas . 2025.
		Robust and Calibrated Detection of Authentic Multimedia Content , Submitted.
		Sarim Hashmi, Abdelrahman Elsayed, Mohammed Talha Alam, Samuele Poppi, Nils Lukas . 2025.
Journal Publications	2	Privacy-Preserving Machine Learning [Cryptography]
	[IEEE S&P'23]	Florian Kerschbaum, Nils Lukas . IEEE Security & Privacy, Volume 21, Issue 6, 2023.
	1	SunFlower: A new Solar Tower Simulation Method for use in Field Layout Optimization
	[AIP'18]	Pascal Richter, Gregor Heimig, Nils Lukas , Martin Frank. AIP Conference Proceedings, Volume 2033, Issue 1, 2018.
Service	Area Chair	
	• AAAI Conference on Artificial Intelligence (AAAI), Senior Program Committee	2027
	• Neural Information Processing Systems (NeurIPS)	2026
	• International Conference on Learning Representations (ICLR)	2026
	• International Conference on Machine Learning (ICML)	2026
	Program Committee	
	• ACM Conference on Computer and Communications Security (CCS)	2025
	• IEEE Symposium on Security and Privacy (IEEE S&P)	2025–2027
	• Recent Advances in Intrusion Detection (RAID)	2024
	Artifact Evaluation Committee	
	• The ACM Conference on Computer and Communications Security (CCS)	2023, 2024
	Reviewer	
	• European Conference on Computer Vision (ECCV)	2026
	• Conference on Language Modeling (COLM)	2026
	• British Machine Vision Conference (BMVC)	2026
	• FAGEN Workshop @ ICML	2026
	• Workshop on GenAI Watermarking (WMark @ ICLR)	2025
	• NETYS	2025
	• ACM TheWebConf (WWW)	2025
	• International Conference on Learning Representations (ICLR)	2024, 2025
	• International World Wide Web Conference (TheWebConf)	2024
	• Recent Advances in Intrusion Detection (RAID)	2023
	• Neural Information Processing Systems (NeurIPS)	2022, 2023
	• International Conference on Machine Learning (ICML)	2022, 2025

	• The Conference on Information and Knowledge Management (CIKM) 2020
	Journal Reviewer
	• Communications of the ACM (CACM) 2025
	Other
	• External Examiner, PhD Dissertation, CISPA Helmholtz Center for Information Security 2026 • Faculty Search Committee, Machine Learning Department at MBZUAI 2025–2026 • Chair for the invited faculty talk program, International Symposium on Trustworthy Foundation Models at MBZUAI 2025 • Admission's Committee, MBZUAI Machine Learning Department 2025 • Student Board Member, Cybersecurity and Privacy Institute 2022–2024 • Session Chair, IEEE Symposium on Security and Privacy (S&P) 2023 • Sub-Reviewer, Proceedings on Privacy Enhancing Technologies (PETS) 2021–2023 • School Advisory Committee on Appointments Liaison, CrySP Lab 2022 • Organization, Workshop on Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWeBMeDA) 2018
Teaching	Instructor, MBZUAI, UAE • ML8509: Collaborative Machine Learning (w/ S. Horvath) 2026 • ML8502: Machine Learning Security (14 weeks) 2025 • ML807: Federated Learning (7 weeks) 2025 • ML818: Emerging Topics in Trustworthy Machine Learning (4 weeks) 2024 Teaching Assistant, University of Waterloo, Canada • CS458/658: Computer Security and Privacy 2020, 2021 • CS246 – Object Oriented Programming 2021 Co-Instructor, RWTH-Aachen, Germany • Course: Data-driven Medicine 2018
Education	University of Waterloo, Canada 2019 – 02/2024 Ph.D. in Computer Science • Advisor: Florian Kerschbaum • Thesis: Analyzing Threats of Large-Scale Machine Learning Systems • Thesis Awards: Top Mathematics Doctoral Prize & Alumni Gold Medal RWTH-Aachen, Germany M.Sc. in Computer Science (<i>w/ Distinction</i>) 2016 – 2018 B.Sc. in Computer Science 10/2012 – 2016
Work Experience	Assistant Professor , MBZUAI, Abu Dhabi, UAE since 08/2024 Research Intern , Royal Bank of Canada, Borealis AI, Toronto 2024 • Vertical Federated Learning, hosted by Kevin Wilson Research Intern , Microsoft Research, Cambridge, UK 2022 • Privacy for Language Models, hosted by Shruti Tople & Lukas Wutschitz Research Assistant , RWTH-Aachen, Aachen 2014 – 2018 Student Researcher , DSA Daten- und Systemtechnik GmbH, Aachen 2016 Software Engineer Intern , A.R. Bayer DSP Systeme GmbH, Düsseldorf 2012